

Competitive Graph Recoloring: Core Lyapunov Argument

1 Notation and system model

Let $G = (V, E)$ be a finite undirected graph, with $n = |V|$ vertices and $m = |E|$ edges. Let $C = \{1, 2, \dots, k\}$ be a finite color set. A coloring, or configuration, is denoted by

$$c = (c_i)_{i \in V} \in C^V,$$

where c_i is the color assigned to vertex i . For a vertex $i \in V$, let $N(i) = \{j \in V : \{i, j\} \in E\}$ be its neighborhood. The local conflict cost of vertex i under configuration c is

$$d_i(c_i, c_{-i}) = \sum_{j \in N(i)} \mathbf{1}\{c_i = c_j\},$$

where c_{-i} denotes the colors of all vertices except i . When the colors of the other vertices are fixed, we also write

$$d_i(\gamma, c_{-i}) = \sum_{j \in N(i)} \mathbf{1}\{\gamma = c_j\}, \quad \gamma \in C.$$

The competitive recoloring dynamics are asynchronous. At each activation, a single vertex i is selected. If there exists a color that strictly reduces its local conflict cost, vertex i changes to a best-response color:

$$c_i^+ \in \arg \min_{\gamma \in C} d_i(\gamma, c_{-i}).$$

The update is accepted only when

$$d_i(c_i^+, c_{-i}) < d_i(c_i, c_{-i});$$

otherwise the configuration is left unchanged. A *step* below means one accepted recoloring move.

Define the global conflict potential by

$$\Phi(c) = \sum_{\{u, v\} \in E} \mathbf{1}\{c_u = c_v\} = \frac{1}{2} \sum_{i \in V} d_i(c_i, c_{-i}).$$

Thus $\Phi(c)$ is exactly the number of monochromatic edges in G .

The global minimization of Φ is computationally nontrivial: for $k \geq 2$, minimizing monochromatic edges is equivalent to maximizing the number of bichromatic edges, i.e., to the MAX- k -CUT objective. Since this problem is NP-hard in general, a distributed best-response rule that is guaranteed to reach a locally stable configuration in finite time is valuable even though it does not claim global optimality.

Lemma 1 (Exact potential drop). *Let c and c^+ be two configurations that differ only at one vertex i . Suppose vertex i changes from color $\alpha = c_i$ to color $\beta = c_i^+$ and strictly reduces its local conflict cost, while $c_{-i}^+ = c_{-i}$. Then*

$$\Phi(c^+) - \Phi(c) = d_i(\beta, c_{-i}) - d_i(\alpha, c_{-i}) \leq -1.$$

Proof. Only edges incident to i can change their conflict status, since all other vertices retain their colors. Hence every edge not in $\{\{i, j\} : j \in N(i)\}$ contributes the same amount to $\Phi(c^+)$ and $\Phi(c)$.

Before the update, the number of conflicting edges incident to i is

$$d_i(\alpha, c_{-i}) = \sum_{j \in N(i)} \mathbf{1}\{\alpha = c_j\}.$$

After the update, the number of conflicting edges incident to i is

$$d_i(\beta, c_{-i}) = \sum_{j \in N(i)} \mathbf{1}\{\beta = c_j\}.$$

Therefore the total change in the number of monochromatic edges is exactly

$$\Phi(c^+) - \Phi(c) = d_i(\beta, c_{-i}) - d_i(\alpha, c_{-i}).$$

By the strict-improvement rule, $d_i(\beta, c_{-i}) < d_i(\alpha, c_{-i})$. Since both quantities are integers, their difference is at most -1 . This proves the claim. $\blacksquare$

Theorem 1 (Finite convergence). *For any finite graph $G = (V, E)$, any finite color set C , and any initial configuration $c(0) \in C^V$, the competitive recoloring dynamics terminate after at most $|E|$ accepted recoloring steps. More precisely, if T denotes the number of accepted recolorings before termination, then*

$$T \leq \Phi(c(0)) \leq |E|.$$

The terminal configuration is a pure Nash equilibrium, equivalently a fixed point of the strict best-response dynamics.

Proof. By Lemma 1, every accepted recoloring decreases Φ by at least one. The potential is integer-valued and nonnegative because it counts monochromatic edges:

$$0 \leq \Phi(c) \leq |E| \quad \text{for every } c \in C^V.$$

Thus a sequence of accepted recolorings can have length no larger than the initial potential value. Consequently,

$$T \leq \Phi(c(0)) \leq |E|.$$

In particular, no infinite sequence of accepted recolorings is possible, so the dynamics terminate in finite time.

At termination, no vertex has a color in C that strictly lowers its local conflict cost. Hence no individual vertex can improve by a unilateral recoloring while the other vertices remain fixed. This is precisely a pure Nash equilibrium for the local conflict-cost game, and equivalently a fixed point of the stated strict best-response dynamics. $\blacksquare$

Remark 1 (Average-case scaling and empirical validation). *Theorem 1 gives a deterministic worst-case upper bound of $|E|$ accepted recoloring steps. The accompanying simulations examine randomized asynchronous updates on sparse Erdős–Renyi and Barabasi–Albert graphs with constant average degree, for which $|E| = \mathcal{O}(|V|)$. Across the reported trials, the observed convergence time is substantially below the worst-case edge bound, typically about $0.12|E|$ accepted recolorings. These data support the interpretation that, under randomized asynchronous updates, the adversarial activation orders needed to approach the worst-case bound are rarely sampled in standard sparse graph ensembles. Consequently, the empirical behavior is closer to $\mathcal{O}(|V| \log |V|)$, and in these experiments nearly linear in the network size, although the formal guarantee used here remains the deterministic edge bound above.*